Comparative Analysis of Pre-GAP and Post-GAP Knowledge Distillation

From Pretrained Image Classifiers to Lightweight ConvNet Students

Josue Airton Lopez Cabrejos
RA: 300867

Luis Alberto Vásquez Vargas
RA: 322989

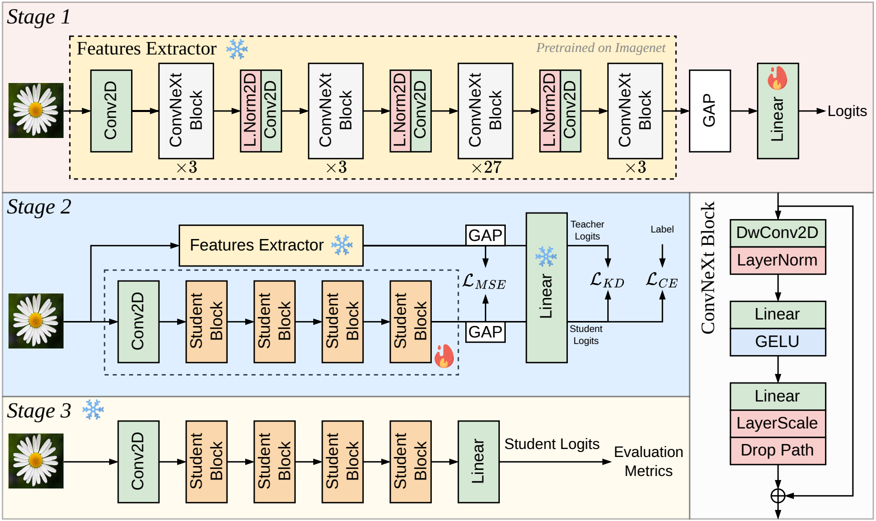
IC – UNICAMP

MO434 – Deep Learning

Instructor: Prof. Dr. Alexandre Xavier Falcão

June 30, 2026

Knowledge Distillation Pipeline



Student Architecture

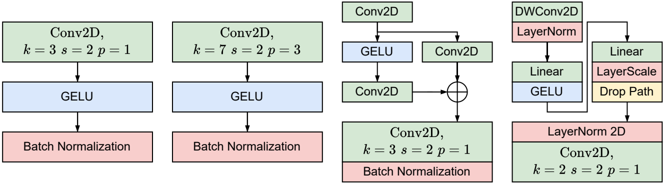


Figure 2: Student architecture-blocks evaluated in the experiments: (a) Student 1, (b) Student 2, (c) Student 3, and (d) Student 4.

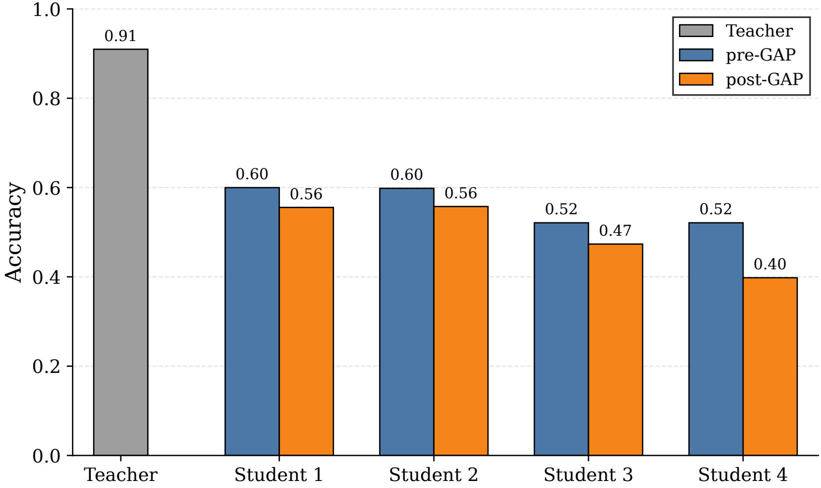
Datasets

Dataset	Task Type
Oxford 102 Flowers	Fine-grained flower classification
Oxford-IIIT Pets	Fine-grained pet classification
Stanford Cars	Fine-grained car classification

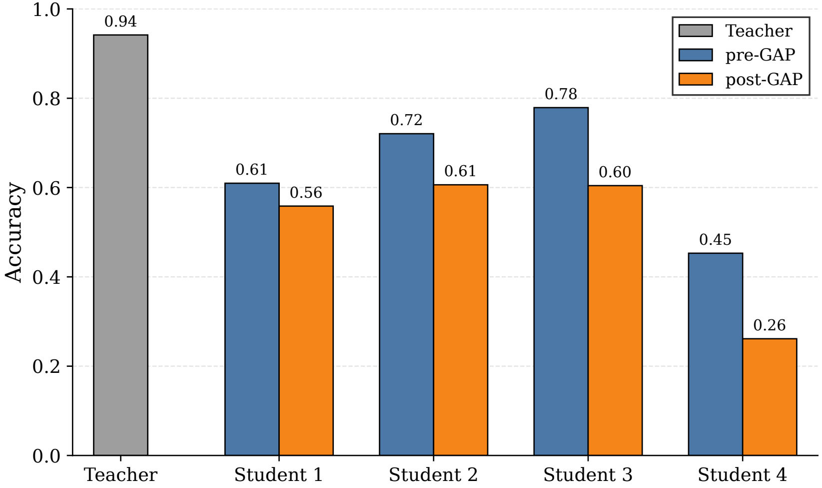
Common Challenge

All datasets require distinguishing visually similar classes using discriminative local and global features.

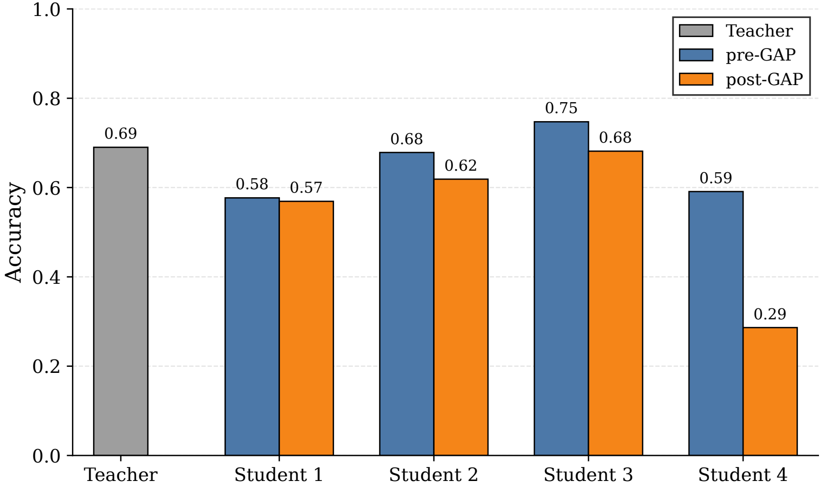
Flowers Accuracy



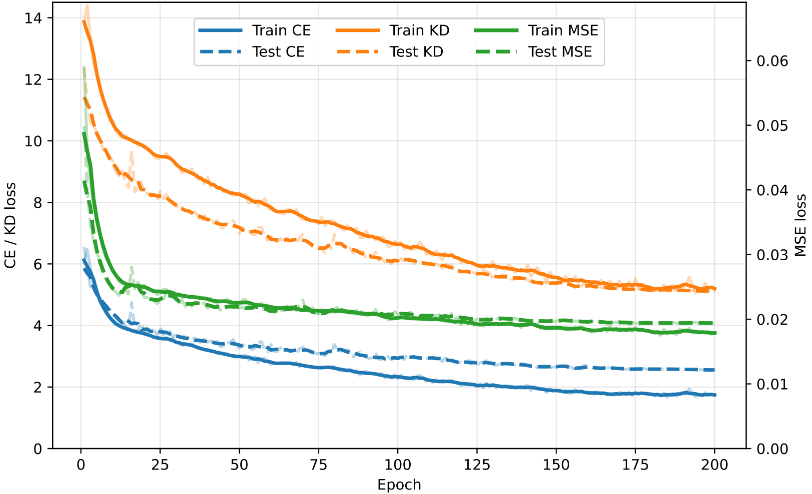
Pets Accuracy



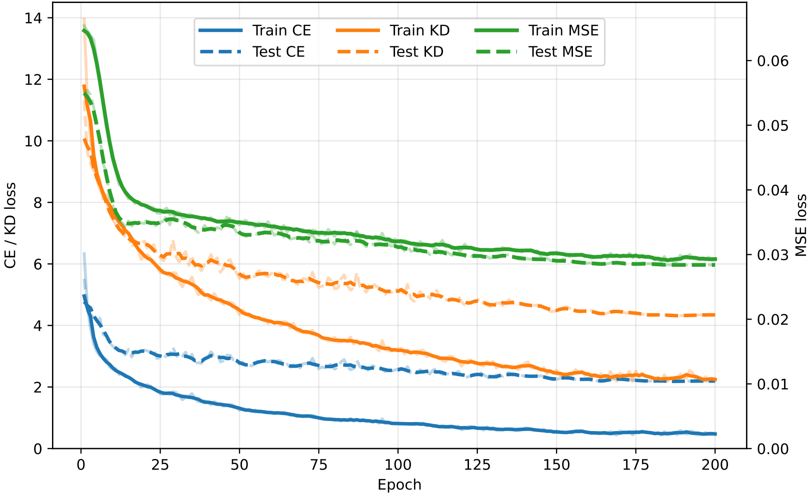
Cars Accuracy



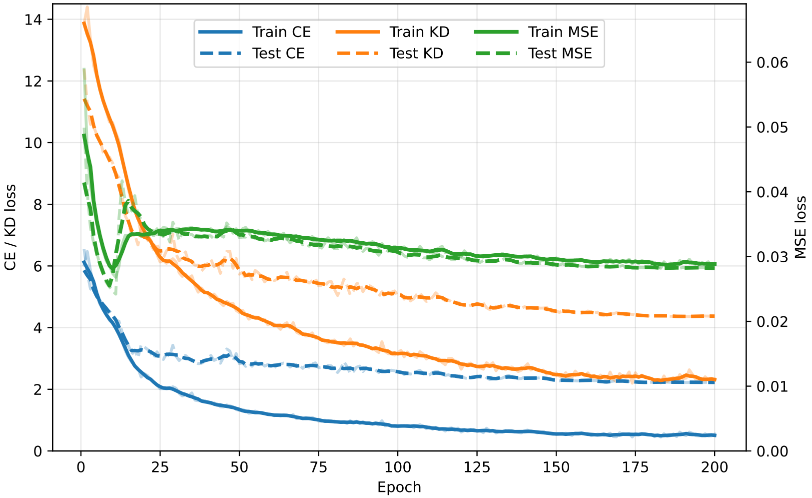
MSE Loss Behavior



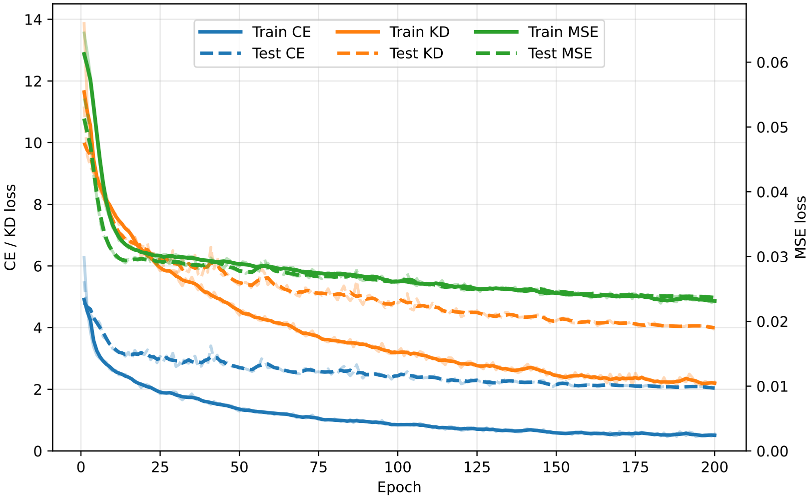
Multi-Loss Behavior



Combined Loss Behavior



Loss Coefficients



Questions?

Thanks! :)

Instituto de Computação

UNICAMP